

Stock Movement Predictor

Developer Visual Reference · XGBoost + SHAP + Streamlit · Idris Foudhaili

1 End-to-end pipeline

The diagram below shows the six stages executed every time the user clicks **Run Analysis** in the Streamlit sidebar.

Stages 1–3 are cached with `@st.cache_resource` so re-running with the same ticker/date combo is instant.

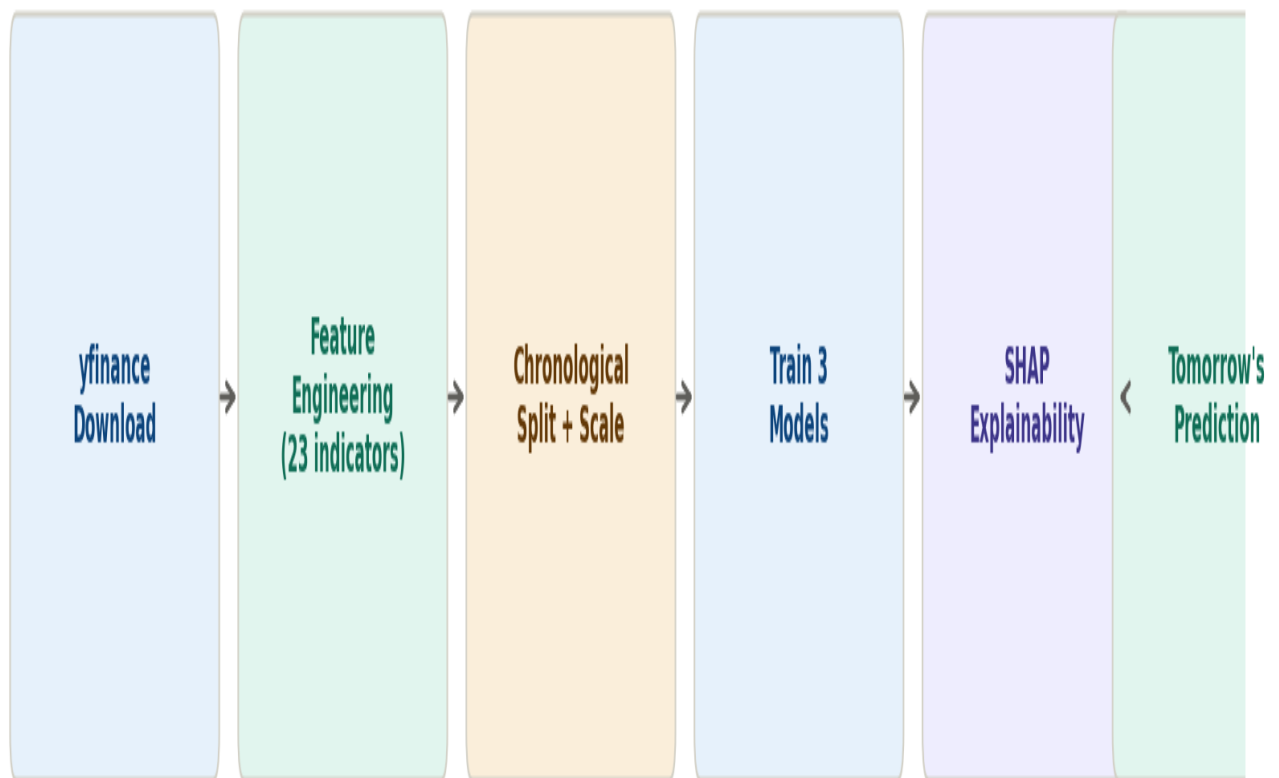


Figure 1 — Pipeline overview. Arrows indicate data flow; each box maps to a module in `src/`.

2 Price action & predicted signals

The main chart (Overview tab) plots the adjusted closing price alongside two overlays: **triangle-up** markers for days where the model predicts a $>+1\%$ move and **triangle-down** markers for $>-1\%$. A dashed SMA-20 line visualises the short-term trend regime.

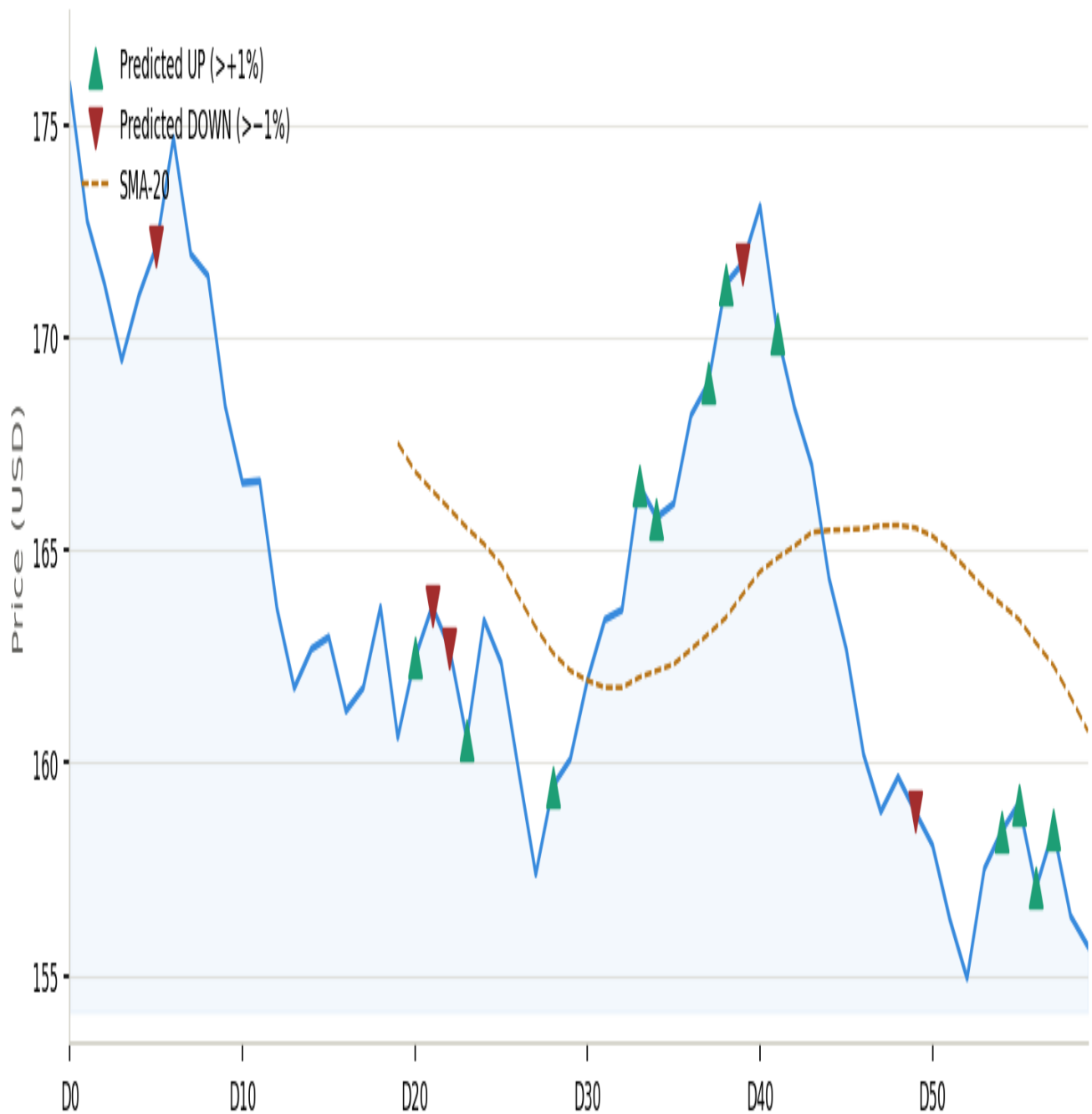


Figure 2 — Simulated 60-day AAPL price with XGBoost signal overlay and SMA-20. UP signals use ▲ (green), DOWN signals use ▼ (red).

Key code reference: `app.py` → `tab_overview` → "Price Action" section · chart drawn with `matplotlib / ax.plot + ax.scatter`.

3 Model accuracy comparison

Three classifiers are trained in sequence. XGBoost consistently outperforms on accuracy while all models hover near 0.50 ROC-AUC — consistent with the Efficient Market Hypothesis. The $\pm 1\%$ move threshold filters noise and focuses on tradeable signals.

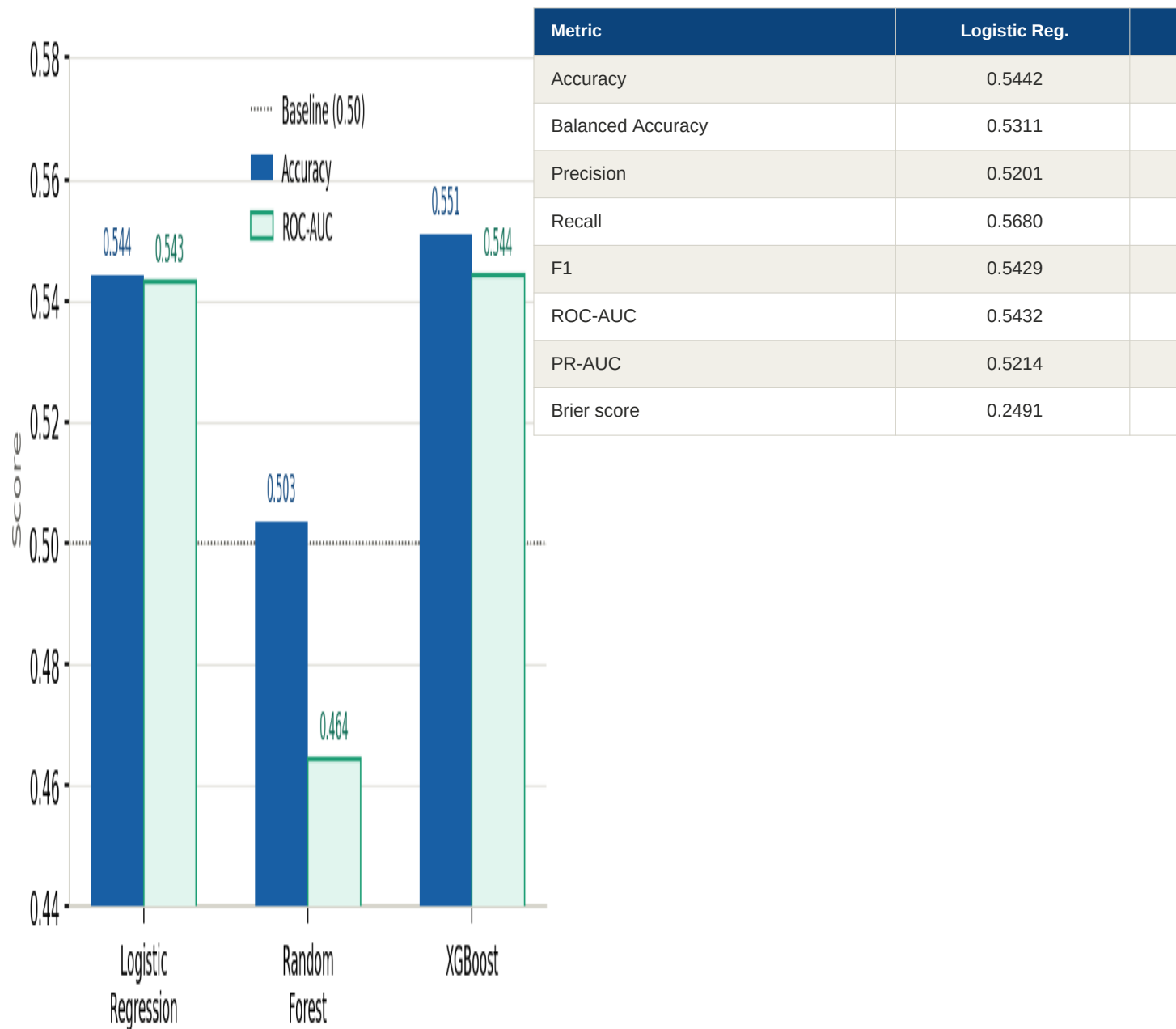
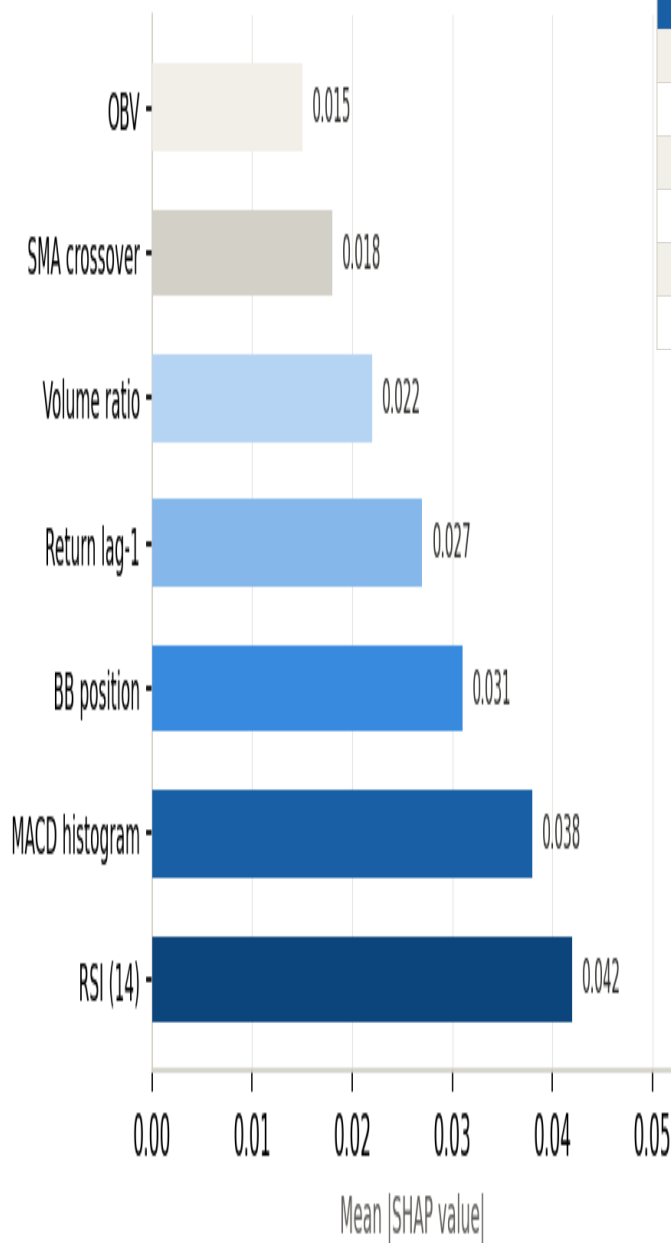


Figure 3 — Model comparison bar chart (left) and full metric scoreboard (right). Blue = Accuracy, pale teal = ROC-AUC.

Key code reference: `src/model.py` → `train_xgboost()` · scores computed in `app.py` → `results_df`.

4 SHAP feature importance (XGBoost)

SHAP TreeExplainer assigns each of the 23 engineered features a mean absolute contribution to the XGBoost output. RSI and MACD histogram dominate, confirming that momentum signals carry the most predictive power for significant moves. The chart maps to the **Explainability** tab.



Category	Features
Moving averages	SMA-20, SMA-50, price_above_sma20/50
RSI	rsi, rsi_overbought, rsi_oversold
MACD	macd, macd_signal, macd_histogram, macd_bullish
Bollinger Bands	bb_upper, bb_lower, bb_width, bb_position
Volume	volume_sma20, volume_ratio, OBV
Lag returns	return_lag{1,2,3,5}, volume_lag{1,2,3,5}

Figure 4 — SHAP mean |value| per feature (left). Feature engineering reference grouped by signal type (right).

Key code reference: `src/features.py` → `engineer_features()` · `src/explainability.py` → `shap.TreeExplainer`.

5 Technical indicators & model evaluation curves

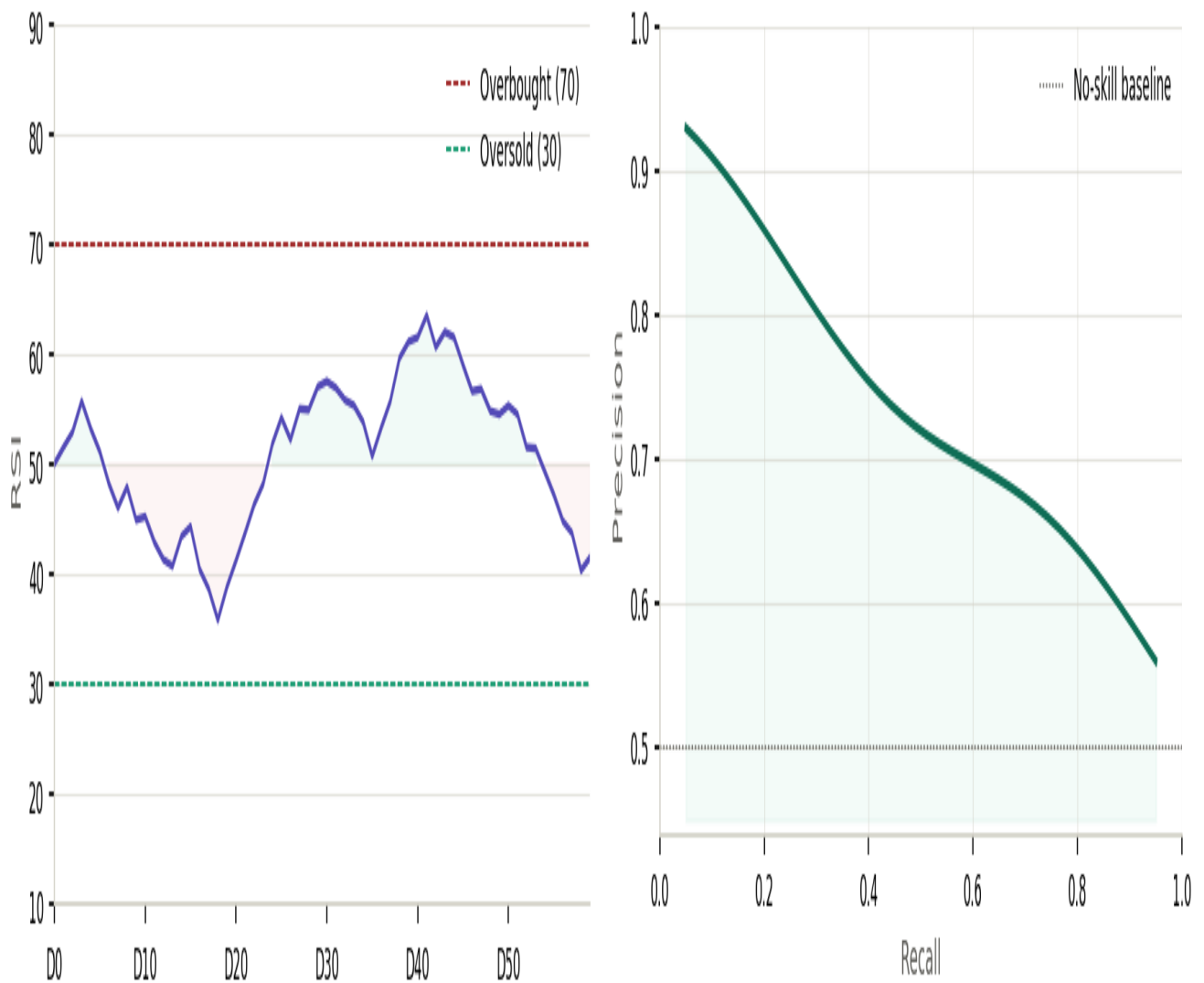


Figure 5 (left) — RSI-14 oscillator. Shaded green above 50 (bullish), red below 50 (bearish). Dashed lines at 70/30 signal overbought/oversold. Figure 6 (right) — Precision-Recall curve for XGBoost. Adjust the threshold slider in the app to navigate the tradeoff.

Threshold slider tip: moving the slider right increases precision (fewer but higher-quality UP signals) at the cost of recall. Default is 0.50; values around 0.55–0.60 often improve signal quality.

6 Quick-start cheatsheet

Action	Command / note
Clone & install	<pre>git clone <repo> && cd stock-predictor python -m venv .venv && .venv\Scripts\activate pip install -r requirements.txt</pre>
Run app	<code>streamlit run app.py</code>
Change ticker	Sidebar → Stock ticker field (e.g. TSLA, MSFT, MC.PA)
Date range	Sidebar → Start / End date pickers (min 2 years recommended)
Tune threshold	Sidebar → Prediction threshold slider (0.30 – 0.70)
Add a feature	Edit <code>src/features.py</code> → <code>engineer_features()</code> → add column name to <code>FEATURE_COLS</code> in

Retrain models	Click Run Analysis again (cache auto-invalidates on new params)
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Foudhaili